An **Interpretable** and **Probabilistic** Approach to **ACS COSI Background** Modeling

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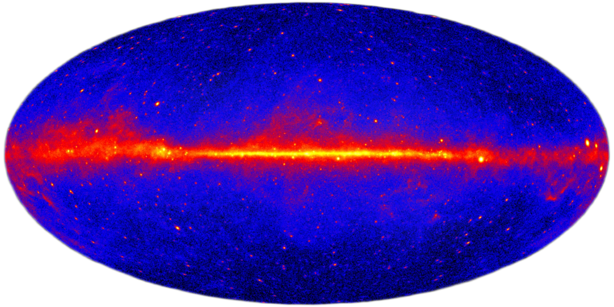
Gamma-Ray Astrophysics

Studies the **most energetic phenomena** in the Universe;
Observes X-ray and **gamma-ray radiation**;

Produced in extreme environments:

- Supernovae
- Compact objects (neutron stars, black holes)



All-sky map in gamma rays: bright regions trace intense high-energy emission. Credit: NASA

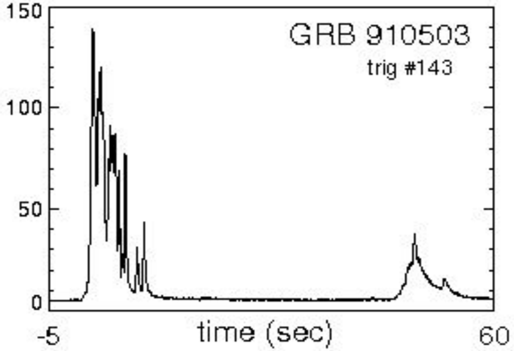
Gamma-Ray Bursts (GRBs)

A **transient** is a **short-lived** astrophysical event (from milliseconds to days).

GRBs are the **most luminous transient** ever known, discovered by *Vela* satellites (1960s).

They matter because:

- Probe extreme physics
- Reveal how compact objects form

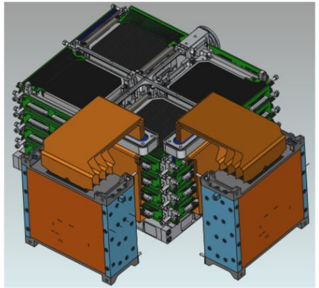


A GRB lightcurve example. Credit: D. Perley

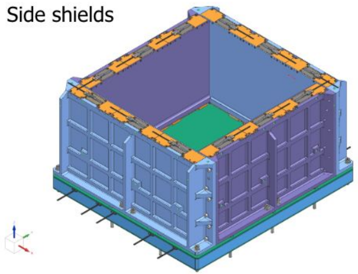
COSI satellite

- The **Compton Spectrometer and Imager (COSI)** is a NASA Astrophysics Small Explorer satellite mission.
- COSI is a soft gamma-ray survey telescope (0.2-5 MeV) **planned** for launch in **2027**.
- Designed to study the **gamma-ray** sky, including high-energy processes in our Galaxy and transient events.
- Provides **improved sensitivity**, resolution, and sky coverage compared to previous missions.

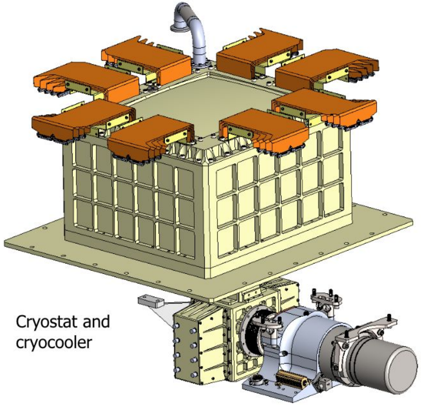
GeDs



Detectors, flex circuits, and read out electronics



BGO



COSI Instrument

Task Definition

Goal

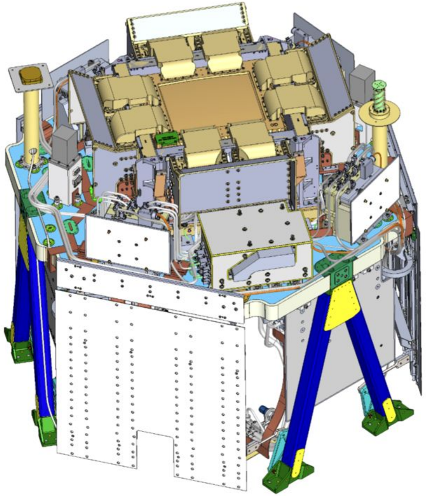
Build an **interpretable** machine learning model to predict the **background** of the **COSI anti-coincidence system** and use it to support **gamma-ray burst** (GRB) detection.

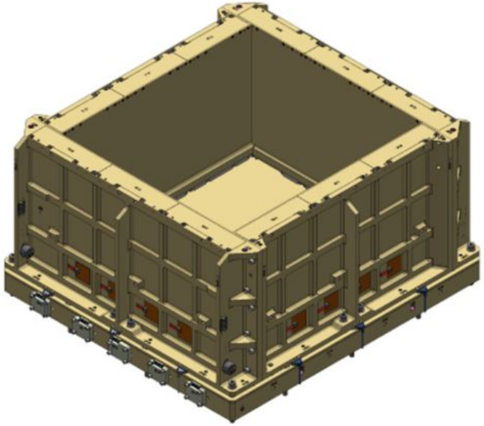
Approach

The background signal is modeled through **conditional probabilistic regression** as a function of the satellite orbital position and attitude.

Interpretability is achieved by combining **post-hoc explanations (SHAP)** with **intrinsically interpretable** models **(Kolmogorov–Arnold Networks)**.

Background predictions are computed **per time bin** and used to define **adaptive thresholds** for transient detection.





Satellite orientation data

→ **x_lat / x_lon**: orientation of the spacecraft +X payload axis, describing the lateral pointing direction of the instrument

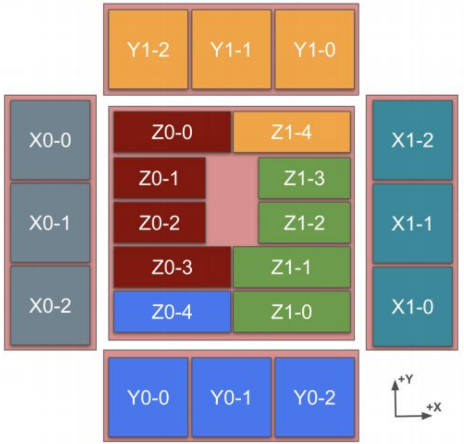
→ **z_lat / z_lon**: orientation of the spacecraft +Z payload axis (boresight), defining the main pointing direction

→ **altitude**: orbital altitude of the spacecraft above Earth's surface

→ **Earth_lat / Earth_lon**: geographic coordinates of the sub-satellite point, encoding the spacecraft position along its orbit

Detectors

COSI consists of **20 BGO scintillator crystals**, grouped into **six detector units** according to their orientation in the spacecraft reference frame.



Z: Two bottom-facing panels

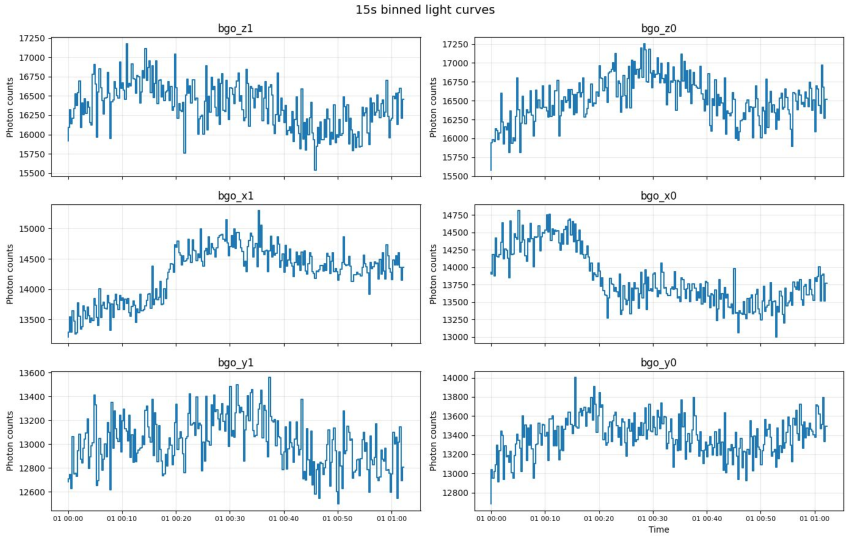
X: Lateral detector panels

Y: Detectors oriented along the Y axis

Scientific Problem | Detector-dependent Background

- The instrumental background is **non-stationary** and **detector-dependent:**

Simulated ACS background (DC3)



Data preparation

Time alignment and interpolation

1. Detector counts are available at **multiple time resolutions** (15 s, 1 s, 50 ms);
2. Spacecraft attitude and orbital metadata are provided at **coarser resolution** (15 s);
3. **Metadata** are **linearly interpolated in time** to match the detector binning;

Feature transformation

1. Angular variables (latitudes and longitudes) are **periodic;**
2. Raw angles are transformed using **sine** and **cosine encoding;**
3. This avoids artificial discontinuities (e.g. at ±180°);
4. Allows the model to focus its **capacity** on meaningful physical dependencies.

Probabilistic Regression for Background Modeling

Input (x)

➔ Spacecraft state and geometry features

Output ($\theta(x)$)

➔ Parameters of the chosen distribution ($\mu(x)$, $\sigma(x)$, ...)

Target variable (y)

➔ Observed photon counts per detector and time bin

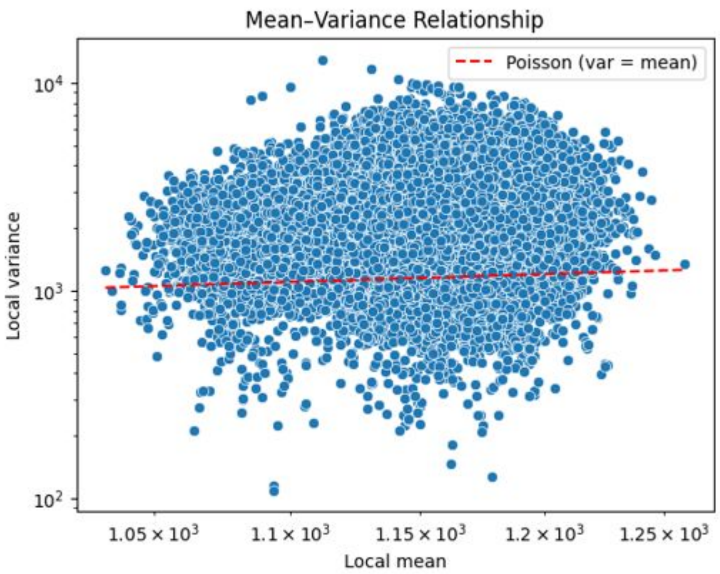
Probabilistic formulation

➔ For **each time bin *t*** and **detector *d*:** $y_{t,d} \sim p(y|x_t; \theta(x_t))$

Training

➔ Minimizing the negative log-likelihood **across all time bins** and **detectors**

Why simple **probabilistic** models **fail**

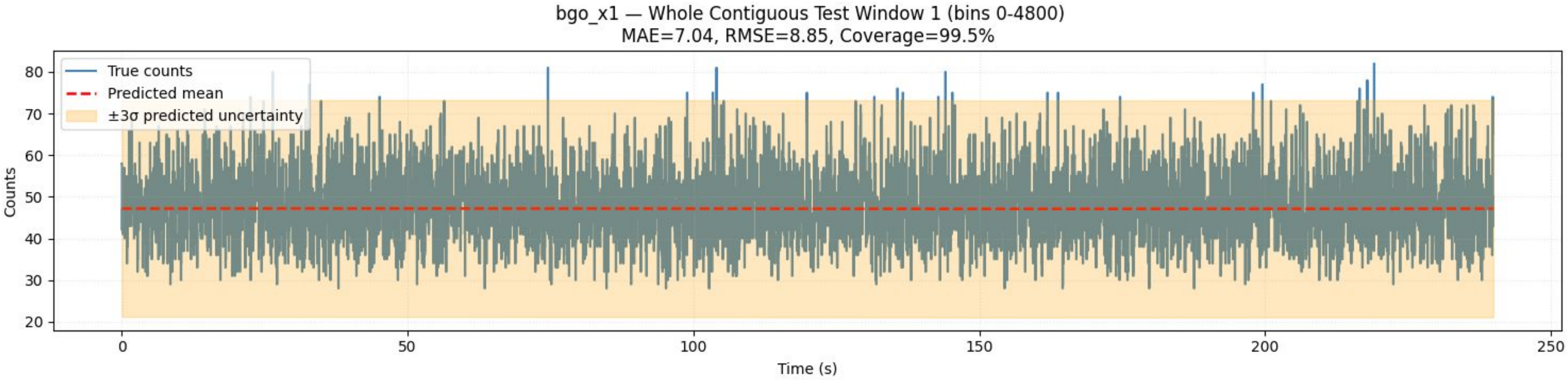


We tried a **Poisson** regression, the **theoretical baseline** for **photon counting, but:**

- ➔ Background counts show strong **overdispersion**
- ➔ **Poisson** assumption fails globally
- ➔ A **single** parametric distribution is **insufficient**

Skew-Normal Regression

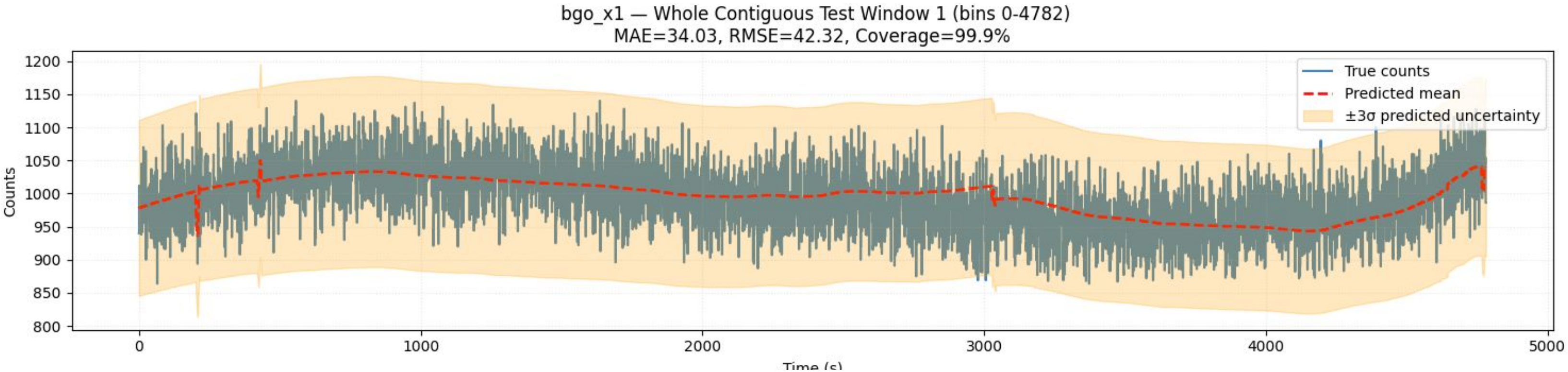
Results at **50 ms time res.** | *4-minute window*



- ➔ Uncertainty bands are calibrated to a **nominal 3σ coverage (~99.7%)** using **Gaussian-equivalent quantiles** (since the 3σ rule does not directly apply to skewed distributions, coverage is enforced via PPF-based quantile matching)

Skew-Normal Regression

Results at **1 s time res.** | *80-minute window*



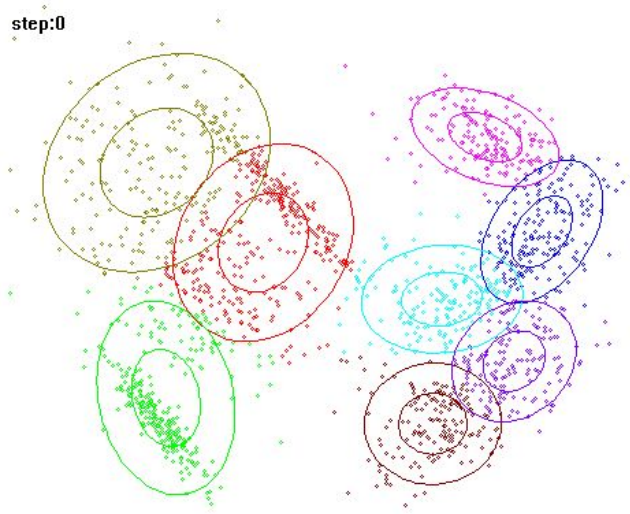
- ➔ High **local variability:** overly **sensitive** to short **fluctuations**
- ➔ The model can be adapted also to **lower time resolution** data, in order to account for transients with a **longer duration** (longer than two seconds).

Gaussian–Mixture Regression (GMR)

The theory behind

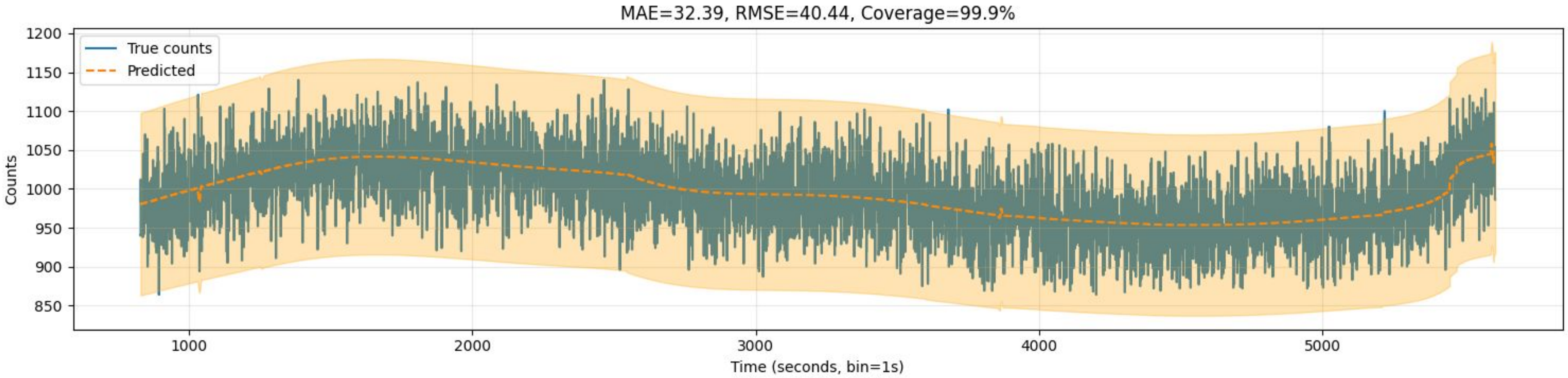
- ➔ **Joint** probabilistic modeling of **inputs** and **target**
- ➔ **Mixture** of **Gaussian components**, which parameters are estimated with the EM algorithm.
- ➔ Allows **multimodal** conditional predictions
- ➔ Classical probabilistic baseline

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$



Gaussian–Mixture Regression (GMR)

Results at **1 s time res.** | *80-minute window*



➔ Gaussian Mixture Regression produces a more **stable** estimate by **averaging** over latent background regimes;

➔ The nominal **3σ coverage (~99.7%)** is **not a strict Gaussian mapping**. For mixtures, coverage is approximated using **symmetric bounds** around the predictive **mean**.

SHAP (**SH**apley **A**dditive ex**P**lanations)

- ➔ SHAP explains a prediction by **decomposing it into additive contributions of each input feature**

- ➔ Contributions are computed **relative to a baseline prediction** (expected model output)

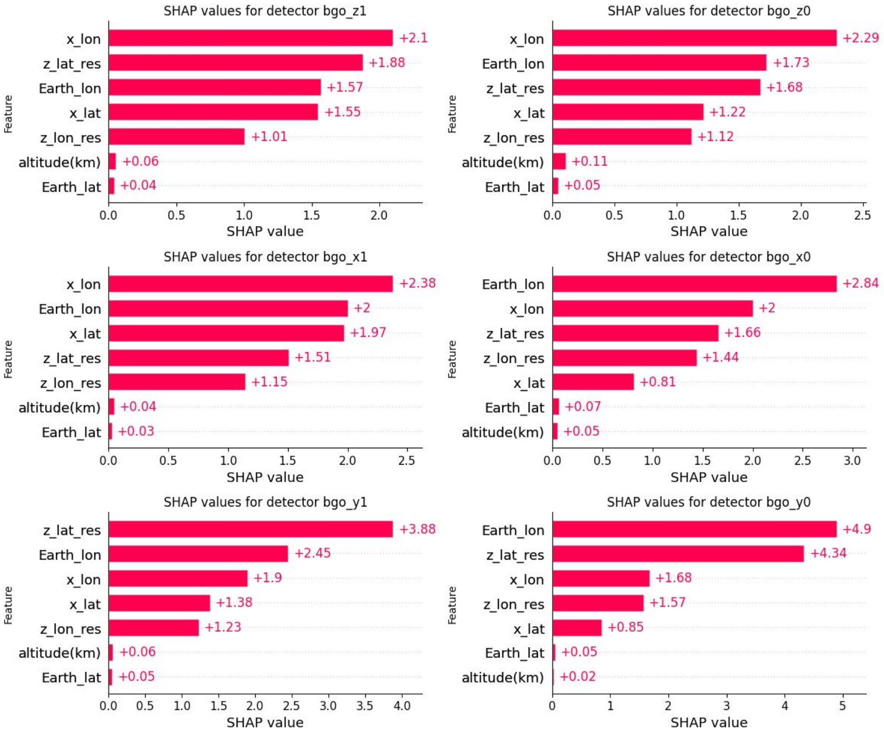
- ➔ Each feature contribution measures **how much it shifts the prediction away from the baseline**

- ➔ Contributions are averaged over **all possible feature orderings**, ensuring fair attribution even with correlated features

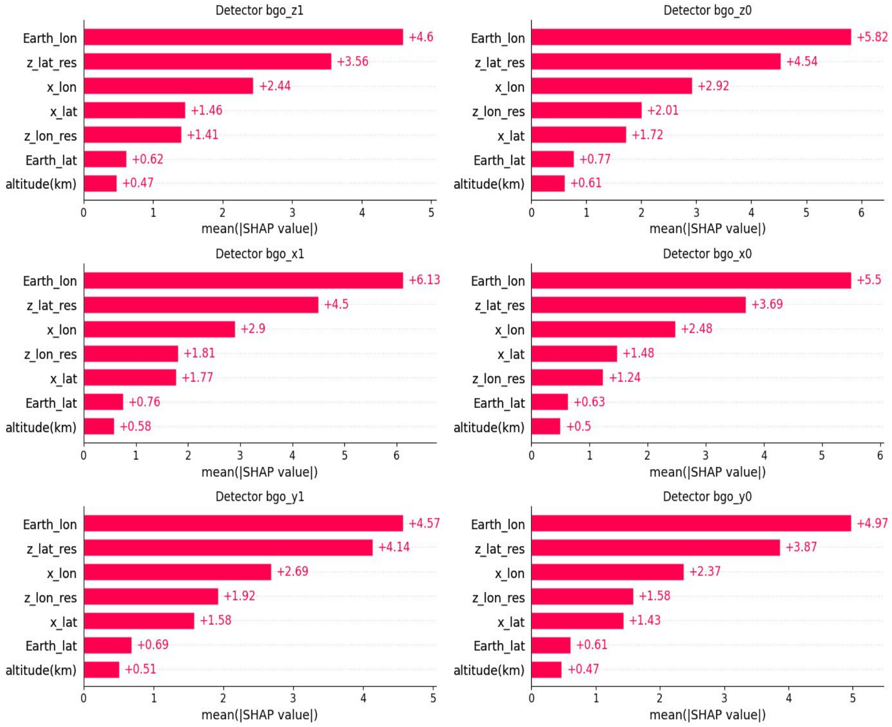
- ➔ **Global explanations:** aggregation of local contributions over the dataset
- ➔ **Model-agnostic:** relies only on model inputs and outputs, not on internal structure

SHAP — feature contribution

GMR model

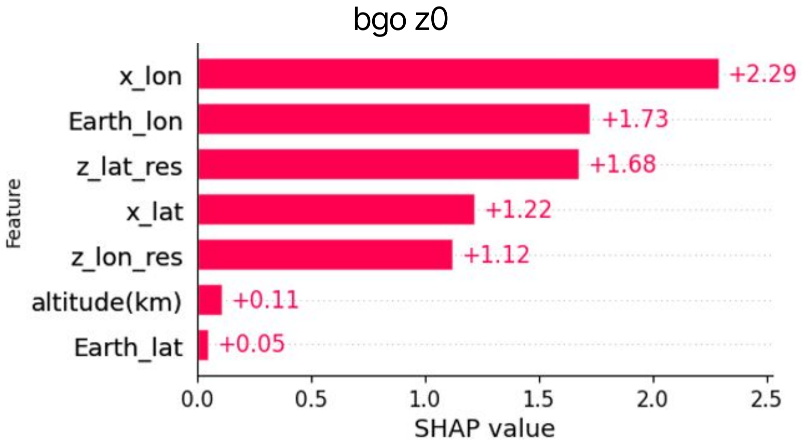


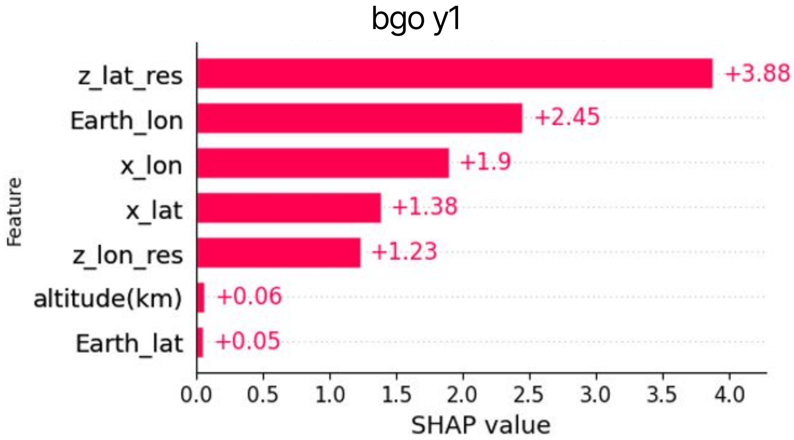
Skew-Normal model



SHAP — feature contribution in detail

GMR model





bgo z0: The dominance of longitudinal features suggests that the z0 detector response is strongly modulated by the spacecraft orientation and its position along the orbit.

bgo y1: Response is strongly driven by the boresight latitude (z_lat_res), indicating a high sensitivity to the spacecraft pointing geometry relative to Earth.

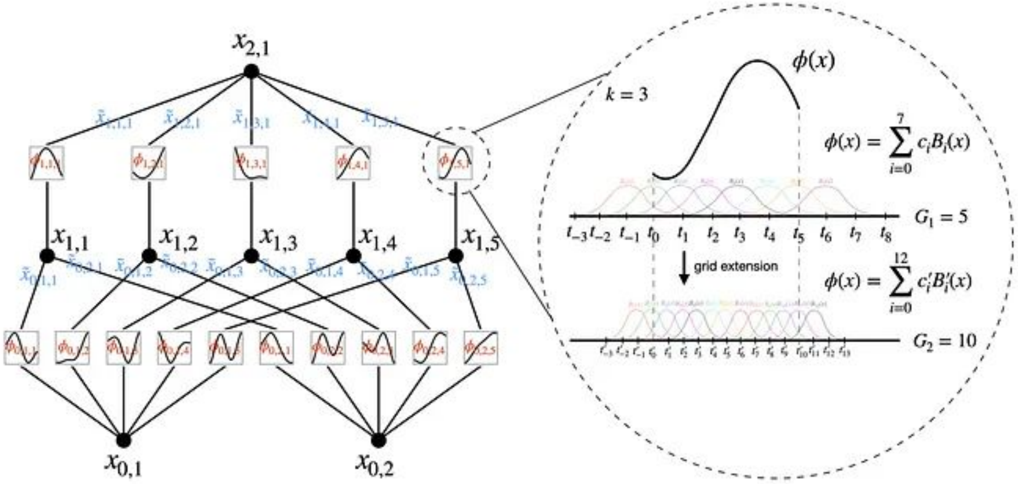
Kolmogorov–Arnold Networks (KANs)

The theory behind — What are KANs?

KANs replace traditional linear weights with **learnable univariate functions** on the network **edges** rather than nodes, as in standard MLPs.

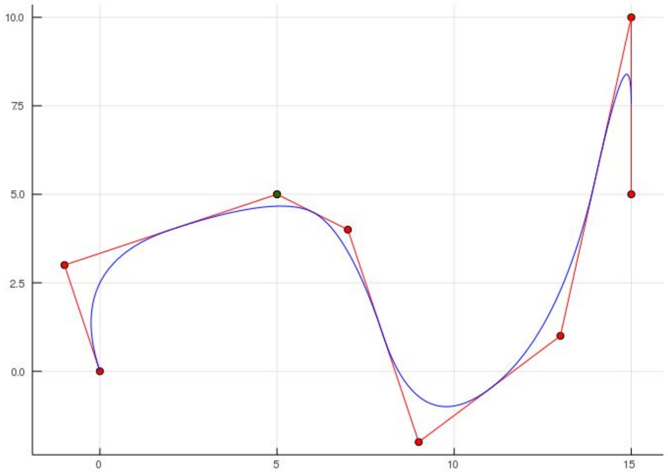
Multivariate mappings are constructed as **compositions** of these functions.

This structure enables explicit and **directly inspectable** functional **relationships**.



Kolmogorov–Arnold Networks (KANs)

The theory behind — B–splines and interpretability



Edge functions are parameterized using **B-splines**, which provide:

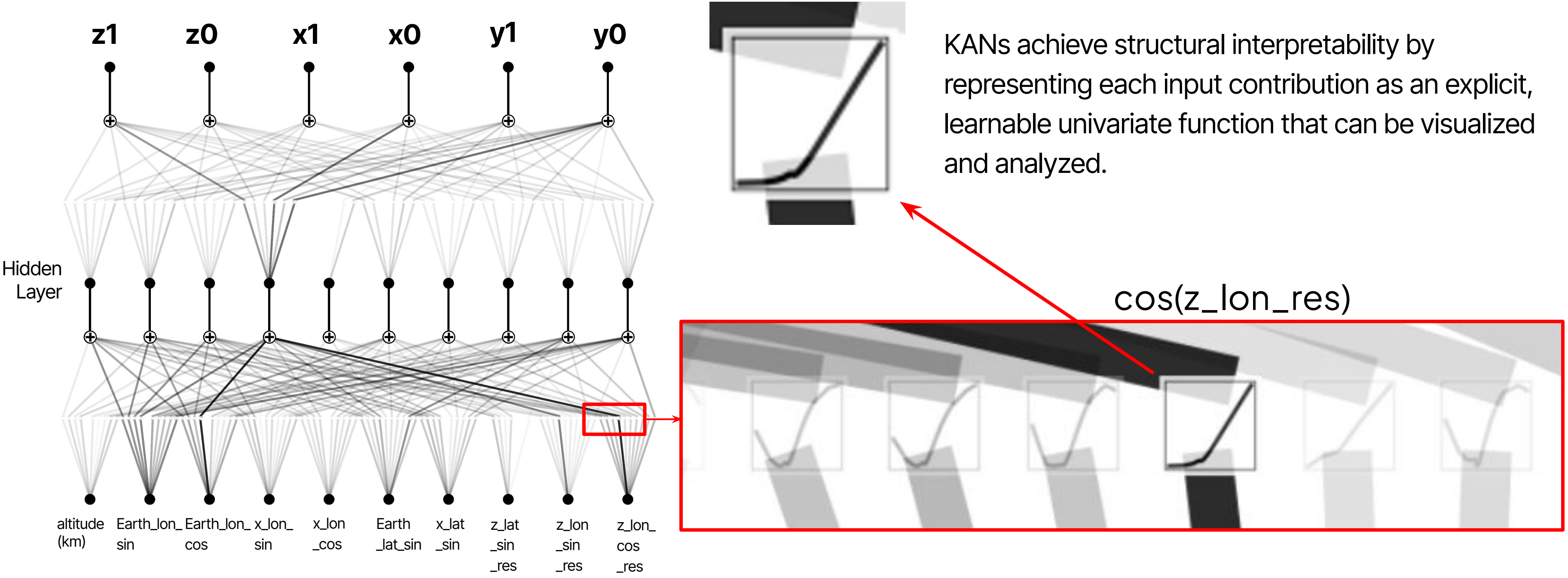
- ➔ Smoothness
- ➔ Locality
- ➔ Controllable complexity

Learned functions can be:

- ➔ Visualized
- ➔ Simplified
- ➔ Approximated by **symbolic** expressions

KAN — interpretability **by design**

Learned univariate functions



KANs achieve structural interpretability by representing each input contribution as an explicit, learnable univariate function that can be visualized and analyzed.

GRBs simulation and **injection**

1. To test the model we used GRB simulated using MEGAlib and Fermi/GBM models;

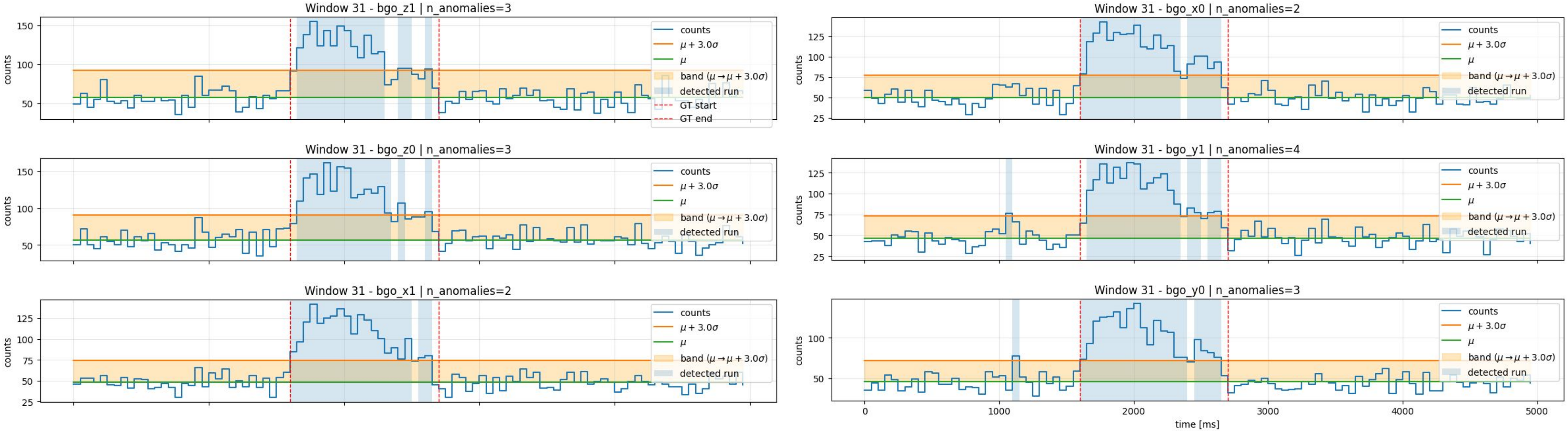
2. GRB signals are injected by **adding simulated GRB counts** to the **background data**, preserving the original background statistics;

3. GRB events are injected at **random** times and windows to provide a simple, model-agnostic **baseline** that tests whether the anomaly detection pipeline can react to generic transient perturbations, without claiming physical realism;

4. Detection is based on **significant positive deviations** from the predicted background distribution;

5. Excesses are identified at **3σ, 4σ,** and **5σ significance levels**, computed from the **model-predicted uncertainty.**

Anomaly Detection | GRB example



➔ Here we can see how the model automatically **flags** a positive deviation from the predicted background, strong enough to exceed the **3σ** threshold. In our pipeline, the detection statistic is the **number** of **time bins above** the **threshold**, which in this case equals **three**.

Final Considerations

➔ We developed a **conditional probabilistic model** to predict the non-stationary background of COSI Anti-Coincidence System;

➔ Background modeling is performed **per detector and per time bin**, capturing both expected counts and uncertainty;

➔ **Interpretability** is a **central design choice**, achieved through model-agnostic explanations (SHAP) and intrinsically interpretable models (KANs);

➔ Feature attributions and learned functions are **consistent with detector geometry** and **physical expectations**, supporting model validity;

➔ Synthetic GRB injection demonstrates a **proof-of-concept use case** for **anomaly detection.**

Thank you!

An **Interpretable** and **Probabilistic** Approach to **COSI ACS Background** Modeling

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INAF Supervisor: A. Bulgarelli
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Why interpretability matters

Background modeling constraints

1. Background variations are driven by **physical and geometrical effects**
2. Detector response is **direction-dependent**
3. Input features are **strongly correlated**

Why interpretability is required

1. Validate model behavior against **physics**
2. Understand **detector-specific** effects
3. Build **trust** in background predictions

Workflow

1. **Data preparation** — Time **binning**, data **interpolation**, feature **transformations,** data **masking** due to **SAA**

2. **Background modeling** — Conditional **probabilistic** regression of detector background

3. **Model interpretability** — Detector-specific and physically grounded interpretability analysis through **SHAP** and **KANs analysis**

4. **Anomaly detection** — Adaptive thresholds and **synthetic GRB injection** using MEGAlib

3. Physical interpretation with SHAP

Consistent physical drivers:
Across all models, SHAP identifies **spacecraft pointing (x/z axes)** and **orbital phase (Earth longitude)** as dominant contributors; **altitude** is always **secondary**.

Effect of angular encoding:
Using **sin/cos** redistributes SHAP across periodic components, improving interpretability by separating **orbital phase** from **pointing direction**, without changing the dominant factors.

Model-dependent behavior:

- **Skew-Normal:** more **local, detector-specific** contributions, reflecting sensitivity to asymmetric background variations.
- **GMR: smoother, more global** attributions, capturing dominant orbital regimes.